Şefika Öztürk

Nashville, TN · sefika.n.ozturk@vanderbilt.edu · 615.484.6056
sefikaozturk.com · github.com/sefikaozturk · linkedin.com/in/sefikaozturk

Education

Vanderbilt University — *Ingram Scholar & Innovation Fellow · Nashville, TN* Expected Dec 2027
B.S. Computer Science & Applied Math; Minor in Data Science

AI Safety & Research

Vanderbilt AI Safety Initiative — *Founder & President · Nashville, TN* Sep 2026 – Present
Founded and lead Vanderbilt's AI safety student organization. Run a 6-person team and a technical AI safety and governance fellowship for 25 fellows, covering alignment, evaluations, interpretability, and AI policy.

Vanderbilt ISIS — *Undergraduate Researcher · Nashville, TN* Fall 2026 – Present
At the Institute for Software Integrated Systems, researching agentic jailbreaking of LLM-powered educational applications with Prof. Meiyi Ma.

Good Start Labs — *AI Research Intern · New York, NY / San Francisco, CA* May 2026 – Aug 2026
Ran RL post-training experiments (LoRA fine-tuning of open base models via Tinker) on game environments, designing reward functions and Inspect-based eval harnesses to measure capability gains on agentic tasks. Reproduced GDPval and found that headline mean scores masked reward-shortcutting: models over-optimized for brevity, improving on short constrained tasks while regressing on long-horizon ones. Built self-play evaluations pitting fine-tuned checkpoints against base and frontier models to separate real gains from metric artifacts; tracked runs in Weights & Biases with cost-budgeted pass@k.

Vanderbilt Data Science Institute — *Generative AI Researcher · Nashville, TN* Jan 2025 – May 2025
Studied bias and fairness in LLMs (fine-tuning, bias detection); built the Tweet Legal Risk Analyzer.

Koç University — *Summer Research Intern · Istanbul, Türkiye* Jun 2021 – Aug 2021
Led a student team on fermionic information theory and quantum biology (Heisenberg models).

Experience

Invariant — *Co-Founder · San Francisco, CA* Mar 2026 – Present
Infrastructure and Neo4j-backed identity layer enabling autonomous agents to discover, call, chain, and pay for external APIs as tools; prototyped agent workflows for reliable multi-step tool use. 1st place, Vanderbilt Firestarter Pitch Competition.

Friends of Shelby Park & Bottoms — *Ingram Scholar Project · Nashville, TN* May 2025 – Jul 2025
Scoped, funded, and shipped an analytics dashboard adopted as the org's primary data tool.

Python Turkish Documentation — *Co-Founder · Antalya, Türkiye* Dec 2021 – Aug 2024
Co-founded the first official Python docs translation group in Turkey; reached ∼300,000 people in launch week.

Antalya ISAS · Unmanned Underwater Systems — *Co-Founder · Antalya, Türkiye* Apr 2021 – May 2023
Co-built tcGrida, a deep-sea robot for 200m dives of 6+ hours; led the live video feed and PyQt app.

Bahçeşehir University Game Laboratory — *Project Manager · Istanbul, Türkiye* Feb 2022 – Oct 2022
At 17, won TechCamp Istanbul and secured U.S. State Department funding to lead Los Problemos, a game teaching disinformation literacy. Managed grants, reporting, and a design/dev team.

Research & Policy Writing

- *The AUTONOMY Act: Liability and Provenance for Autonomously Generated Deepfakes*. Policy memo to the Senate Commerce Committee, co-authored with Rovier Chen (2026).
- *Utilizing Augmented Reality to Improve Geometry Education*. TÜBİTAK 3rd nationally, 1st regionally.
- *Fermionic ToolQIT: Customised Functions for Fermionic Quantum Information Theory*.
- *Heisenberg XX Chain Model for Quantum Effects in D. Radiodurans Nitric Oxide Synthesis*.

Selected Projects

- **Polytopia** — Multi-agent environment for RL and mechanistic interpretability. HackMIT 2025 Grand Prize.

Honors & Awards

- **Ingram Scholarship** — Vanderbilt's most selective merit scholarship; top 1% of applicants.
- **HackMIT 2025 Grand Prize** (3rd of 319 projects) · **Google Solutions Challenge 2025** North America Winner · **Vanderbilt Firestarter 2026** 1st Place.
- **TÜBİTAK National Math Research Competition** — 3rd nationally, 1st regionally (Turkey's ISEF).

AI Safety Training & Skills

Training: Global Challenges Project; in-person AI Safety Bootcamp; BlueDot AI Safety and Technical AI Safety courses.
Technical: Python, PyTorch, Hugging Face, RL post-training (SFT/GRPO, LoRA/Tinker), Inspect, W&B.